AI-DRIVEN COOPERATIVE PERCEPTION AND DIGITAL TWIN PLATFORM FOR CONNECTED AUTONOMOUS VEHICLES
PHASE I REPORT

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING





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September 2026

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BONAFIDE CERTIFICATE

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ABSTRACT

Connected Autonomous Vehicles (CAVs) rely on onboard sensors such as cameras, LiDAR, radar, and GPS to provide situational awareness and enable driving decisions, but each has limitations such as blind spots, occlusion, reduced sensing ranges, and poor performance in certain weather conditions. Additionally, present vehicular perception systems operate in isolation and have little ability to share environmental information or maintain a cooperative digital representation of the driving environment.

This project will demonstrate an AI-Driven Cooperative Perception and Digital Twin Framework for Connected Autonomous Vehicles (CAVs), building upon Artificial Intelligence, Computer Vision, Cooperative Perception, and Digital Twin areas to facilitate environmental comprehension and improve driving decisions. The proposed framework will employ an AI powered perception system to identify relevant traffic objects and have a cooperative module for connected vehicles to share structured information. A digital twin will be maintained in the cloud to model and maintain a virtual real-time connected vehicle and process events and analytics for future intelligent transportation systems applications. The proposed solution will contribute to smart mobility by improving situational awareness, promoting road safety, and providing a scalable framework for future connected and autonomous vehicle platforms.

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavour to put forth this report. Our sincere thanks to our Chairman Mr. S. MEGANATHAN, B.E, F.I.E., our Vice Chairman Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S., and our respected Chairperson Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D., for providing us with the requisite infrastructure and sincere endeavouring in educating us in their premier institution.

Our sincere thanks to Dr. S.N.S MURUGESAN, M.E., Ph.D., our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to Dr. Manoranjini, Ph.D., Professor and Head of the Department of Computer Science and Engineering for her guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, Dr. P. Muneeshwari, Associate Professor, Department of Computer Science and Engineering for his valuable guidance throughout the course of the project. We thank our Project Coordinator, Mr. Karthik, B.E., M.E., Assistant Professor, Department of Computer Science and Engineering for his useful tips during our review to build our project.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CAV	Connected Autonomous Vehicle
DT	Digital Twin
OSM	OpenStreetMap
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything
CARLA	Car Learning to Act
ITS	Intelligent Transportation System
GPS	Global Positioning System
API	Application Programming Interface
IoT	Internet of Things
BEV	Bird's-Eye View

CHAPTER 1

INTRODUCTION

Connected and Intelligent Transportation Systems are emerging as an important technology for improving the safety, efficiency and coordination of modern transportation networks. Connected vehicles are capable of exchanging information such as their position, speed, route status and surrounding conditions with other vehicles or transportation infrastructure.

In a conventional transportation environment, individual vehicles primarily depend on their locally available information. As a result, one vehicle may have limited knowledge about the current state or operating conditions of another vehicle travelling on the same road network. This limitation becomes important in urban transportation systems where multiple buses operate on shared or intersecting routes.

Cooperative perception provides a mechanism through which connected vehicles can share relevant information with one another. Instead of considering each vehicle as an isolated entity, vehicles can contribute information to a common representation of the transportation environment.

A Digital Twin provides a virtual representation of a physical system and its operating environment. In the context of connected vehicles, a Digital Twin can maintain the virtual state of vehicles and provide a synchronized representation for monitoring and analysis.

The proposed project combines these concepts in a simulated urban bus environment. Two buses are simulated using CARLA within an urban road network created using OpenStreetMap. Their operational information is generated and exchanged, and the resulting information is represented using a Digital Twin.

1.1 GENERAL

The project is designed around the idea that a connected transportation environment should not depend only on isolated vehicle observations. A bus operating on one part of a road network can contribute useful information to another bus that is approaching the same area. This shared information can improve the common understanding of the transportation environment.

OpenStreetMap is used as the source for realistic urban road-network information. CARLA provides the simulation environment in which the road network and vehicles can be represented. The combination allows the project to demonstrate connected-vehicle concepts without requiring physical autonomous buses.

The Digital Twin layer maintains a virtual representation of the simulated vehicles. Rather than simply displaying isolated sensor values, the virtual representation can be used to observe the state of multiple vehicles together.

The current project is intentionally focused on a manageable two-bus scenario. It demonstrates the core concepts of vehicle-state generation, information sharing and Digital Twin representation while leaving large-scale fleet deployment and physical vehicle integration for future development.

The system therefore serves as a prototype framework for studying cooperative transportation information and Digital Twin synchronization in an urban environment.

1.2 OBJECTIVE

The primary objective of the project is to design and implement a simulation-based platform that demonstrates cooperative information sharing and Digital Twin representation for connected urban buses.

- Model a realistic urban road environment using OpenStreetMap data.

- Simulate two connected urban buses using CARLA.
- Continuously generate important vehicle-state information such as position, speed and route status.
- Enable exchange of operational information between the two simulated buses.
- Maintain the current state of both buses in a Digital Twin.
- Provide a synchronized virtual representation of the connected transportation environment.
- Demonstrate improved situational awareness through shared vehicle information.
- Provide a foundation for basic transportation monitoring and decision support.

1.3 EXISTING SYSTEM

Existing urban transportation systems commonly provide individual vehicle tracking and route monitoring. Information such as vehicle position, speed and route status may be available to transportation operators or individual vehicles. However, these systems may treat vehicle information separately rather than maintaining a continuously synchronized representation of all connected vehicles.

Two buses operating on the same road network may independently know their own position and speed but may not continuously share this information through a common virtual representation. This limits the ability of one vehicle to benefit from information already observed by another vehicle.

- Limited awareness of the state of other buses.
- Lack of continuous vehicle-to-vehicle information exchange.
- Separate vehicle-state representations.
- Absence of a synchronized Digital Twin for the connected environment.

- Reduced coordination between vehicles operating on related routes.
- Limited common situational awareness for transportation monitoring.

1.4 PROPOSED SYSTEM

The proposed system develops a Digital Twin-based connected-vehicle framework for two urban buses. The system first creates a realistic urban road environment using OpenStreetMap data and represents this environment in CARLA.

Two buses are then introduced into the simulated environment. During simulation, each bus generates operational information including its position, speed, route status and current state. This information is made available through a vehicle-to-vehicle communication layer.

The Digital Twin acts as the virtual representation of the connected buses. It receives the available vehicle information and maintains a synchronized view of the simulated transportation environment. The resulting representation supports monitoring, situational awareness and basic transportation decision support.

The current prototype demonstrates vehicle simulation, vehicle-state generation and Digital Twin representation. The communication mechanism is currently represented as a prototype or mocked interface, which can be replaced by an actual V2V communication protocol in future work.

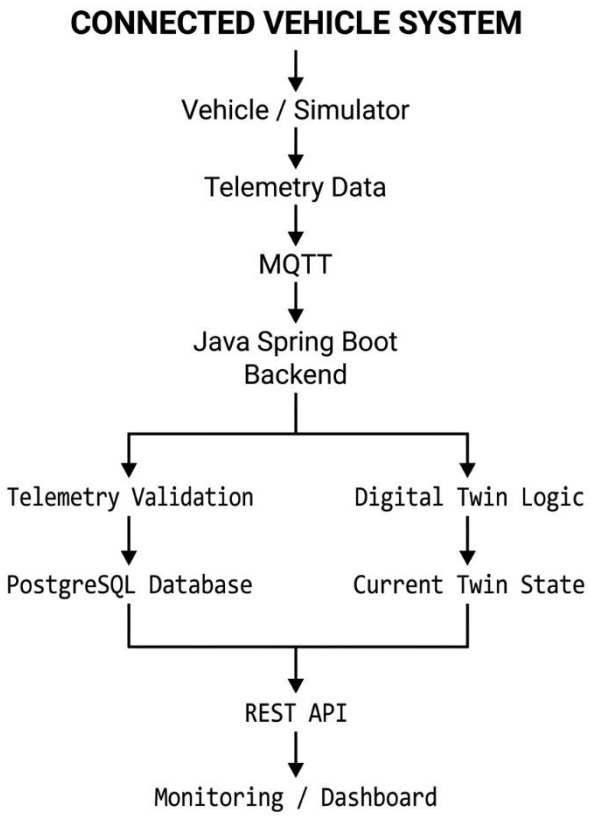


Figure 1.1 Overall proposed project architecture

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Author(s) / Year	Title of the Paper	Methodology / Approach	Inference
Schwarz & Wang, 2022	The Role of Digital Twins in Connected and Automated Vehicles	Review of Digital Twin concepts and their applications in CAVs	Shows how Digital Twins can connect physical vehicles with virtual models for testing, monitoring and development.
Huiyuan Xiong, Zhijun Wang, Guohui Wu, Yuelong Pan, 2022	Design and Implementation of Digital Twin-Assisted Simulation Method for Autonomous Vehicle in Car-Following Scenario	Digital Twin-assisted AV simulation with physical and virtual vehicle interaction	Demonstrates that Digital Twins can support safer and more effective autonomous-vehicle simulation and validation.
Bassel Hakim, Hatem Sarour, Mohamed S. Hefieda, Waleed S. Alasmary, Khaled H. Almotairi, 2023	CCPAV: Centralized Cooperative Perception for Autonomous Vehicles Using C-V2X	Centralized cooperative perception using C-V2X communication	Shows how connected vehicles can share information to improve awareness while addressing communication-resource limitations.
Hieu X. Ngo, Hua Fang, Honggang Wang, 2023	Cooperative Perception With V2V Communication for Autonomous Vehicles	V2V communication with BEV-based perception	Demonstrates that V2V information sharing can help overcome occlusion and improve vehicle perception.
Chenguang Liu, Yunfei Chen, Jianjun Chen, Michael Riley, and Shuang-Hua Yang, 2023	Cooperative Perception with Learning-Based V2V Communications	V2V communication, feature fusion and CNN-based compression	Shows that cooperative perception can improve perception while reducing communication bandwidth requirements.
Huang et al., 2023	Vehicle-to-Everything Cooperative Perception for Autonomous Driving	Survey of V2X cooperative perception, data alignment and fusion	Identifies V2X communication as an important way to extend perception range and improve situational awareness.

Xiang et al., 2023	HM-ViT: Hetero-Modal Vehicle-to-Vehicle Cooperative Perception with Vision Transformer	Multi-agent V2V perception using heterogeneous sensors and Vision Transformers	Shows how vehicles with different sensor types can collaboratively improve 3D object perception.
Scalable Cooperative Perception, 2023	Scalable Cooperative Perception for Connected and Automated Driving	Cooperative perception message-generation and communication analysis	Highlights communication overhead as an important challenge when scaling cooperative perception.
Wang et al., 2024	Smart Mobility Digital Twin Based Automated Vehicle Navigation System: A Proof of Concept	Digital Twin integrated with autonomous-vehicle navigation and V2X	Demonstrates the feasibility of using Digital Twin for smart mobility and automated vehicle navigation.
Zimmer et al., 2024	TUMTraf V2X Cooperative Perception Dataset	Multi-modal roadside + onboard sensor dataset and cooperative 3D perception	Provides data and a benchmark for cooperative perception using vehicle and infrastructure sensors.
Campolo et al., 2024	An Edge-Based Digital Twin Framework for Connected and Autonomous Vehicles: Design and Evaluation	Edge-based Digital Twin architecture for CAVs	Explores an edge-based architecture to support low-latency Digital Twin services for connected vehicles.
Keshu Wu, Pei Li, Yang Cheng, Steven T. Parker, Bin Ran, David A. Noyce, Xinyue Ye, 2024	A Digital Twin Framework for Physical-Virtual Integration in V2X-Enabled Connected Vehicle Corridors	C-V2X + Digital Twin + CARLA simulation	Closely matches the project by synchronizing vehicle/traffic information between physical and virtual transportation environments.
Yucheng Sheng, Hao Ye, Le Liang, Shi Jin, Geoffrey Ye Li, 2024	Semantic Communication for Cooperative Perception Based on Importance Map	Semantic communication and intermediate feature fusion	Reduces unnecessary data transmission while maintaining useful cooperative perception information.
Ahmad Sarlak, Hazim Alborzgan, Sayed Pedram Haeri Bojnordi, Abolfazl Razi, Rahul Amin, 2024	Enhanced Cooperative Perception for Autonomous Vehicles Using Imperfect Communication	Cooperative perception with constrained V2V communication; CARLA simulation	Shows that cooperative perception can improve detection even under communication limitations.
Noé Zapata, Gerardo Pérez-González, Pablo Bustos, Sergio Barroso, Pedro Núñez, 2025	An Efficient and Versatile Digital Twin Model Implementation for Autonomous Connected Vehicles Based on Distributed, Low-Latency Working Memory	Distributed Digital Twin architecture + CARLA + real vehicle validation	Demonstrates a Digital Twin architecture capable of synchronization, simulation and real-time decision support for CAVs.

Parinaz Babaei, Nosrat Riahinia, Omid Mahdi Ebadati E, Ali Azimi, 2025	Towards a Data-Driven Digital Twin AI-Based Architecture for Self-Driving Vehicles	Data-driven DT architecture combining physical, virtual and communication planes	Proposes an AI-oriented Digital Twin architecture for self-driving vehicles using V2V, V2I and cloud connectivity.

Table 2.1 Comparison of Reviewed Literature

2.2 COMPARISON AND DISCUSSION

Aspect	Existing Approaches	Proposed System
Primary focus	Often address cooperative perception, Digital Twin synchronization, or simulation as individual research themes.	Integrates connected-vehicle information exchange and Digital Twin representation in one two-bus urban simulation.
Vehicle scenario	Many studies use autonomous cars, multi-agent settings, datasets or specialized scenarios.	Focused on two connected urban buses operating on a realistic OpenStreetMap-based road network in CARLA.
Communication	V2V/V2X or C-V2X methods may be evaluated with communication constraints.	Current Phase I communication is a prototype/mocked interface designed for future V2V/V2X integration.
Virtual representation	Digital Twin studies emphasize physical-virtual synchronization and monitoring.	Digital Twin maintains a common virtual state of both simulated buses and their environment.
Simulation	CARLA and other simulation environments are used for validation of individual techniques.	CARLA is used as the operational simulation layer and supports repeatable urban-bus scenarios.
Extensibility	Advanced perception, larger fleets, edge/cloud and real communication are common directions.	Architecture is modular and can be extended to larger fleets, real V2V/V2X, sensors, cloud and edge deployment.

Table 2.2 Comparison of Existing Approaches and Proposed System

2.3 RESEARCH GAP AND CONCLUSION

The literature shows that cooperative perception and Digital Twin technologies address complementary limitations. Cooperative perception improves the information available to an individual vehicle by allowing information exchange. Digital Twins provide a structured virtual representation that can be used for monitoring, analysis and synchronization.

Most cooperative-perception studies focus on perception quality, object detection, communication efficiency or sensor fusion. Digital Twin studies focus more strongly on physical-virtual synchronization, simulation and system monitoring. The proposed project connects these research directions through a focused urban-bus scenario.

The project does not claim to implement all advanced techniques discussed in the literature. Instead, the literature provides the theoretical basis for the prototype architecture and identifies future directions such as real V2X, multi-sensor cooperative perception, large-scale fleet simulation and cloud-based Digital Twins.

CHAPTER 3

SYSTEM DESIGN

3. SYSTEM DESIGN

The system design defines how the simulated transportation environment, connected buses, communication layer and Digital Twin interact. The architecture is divided into environment modelling, vehicle-state generation, cooperative communication, Digital Twin and monitoring layers.

OpenStreetMap supplies realistic road-network information. CARLA converts the road environment into a simulation in which two buses can operate. The vehicle-state module continuously updates the operational state of each bus. The communication layer shares information between the buses. The Digital Twin receives this information and maintains a virtual representation that can be monitored.

The design is modular so that the current mocked communication interface can later be replaced with an actual V2V implementation.

3.1 SYSTEM ARCHITECTURE

The proposed system architecture is divided into multiple connected layers so that vehicle simulation, communication, validation, Digital Twin processing, storage, analytics, experiments and monitoring can operate as separate but coordinated modules. The architecture begins with the simulated buses and ends with a monitoring dashboard that provides a common view of the transportation environment. This layered structure also makes each stage easier to test and extend in later phases.

The Simulation and Scenario Layer represents the operational environment. OpenStreetMap provides realistic urban road-network information, while the simulator represents the roads, buses and movement of the vehicles. Bus 1 and Bus 2 continuously produce vehicle-state and scenario information. Typical data includes vehicle identity, position, speed, route status and event information. Scenario conditions such as normal operation, sudden braking, heavy traffic, road obstruction or other events can be used to observe system behaviour.

The Telemetry and Communication Layer transfers the generated information. A Python telemetry publisher collects vehicle and event data, creates structured messages and publishes them to MQTT topics. The Mosquitto broker provides the messaging interface between publishers and subscribers. The design supports bidirectional communication and can later carry real V2V or V2X messages. Controlled network conditions such as latency, packet loss, duplicate or out-of-order messages, stale messages and temporary disconnection can be introduced through experiment configuration.

The Backend Application Layer is implemented using Java and Spring Boot. The MQTT client receives telemetry messages and passes them to the Message Validation Engine. Validation covers message structure, timestamps, sequence information, duplicates, ordering and basic physical plausibility. Valid data is used by the Digital Twin Core to maintain the latest and historical state of each bus. Freshness and Reliability Engines assess whether updates are current and consistent. The Adaptive Synchronization Engine can use priority, event severity, freshness risk and reliability risk to determine appropriate update behaviour.

The Data Persistence Layer uses PostgreSQL to retain vehicle information, high-frequency vehicle states, Digital Twin history, events, communication records, synchronization records and experiment results. The Analytics and Metrics Layer can calculate position and velocity errors, information age, reliability metrics, communication overhead, event propagation delay, stale-state percentage and recovery time. The Experiment and Configuration Layer defines scenario type, duration, network conditions, synchronization

rates, event conditions and reproducibility settings, allowing baseline and adaptive approaches to be compared.

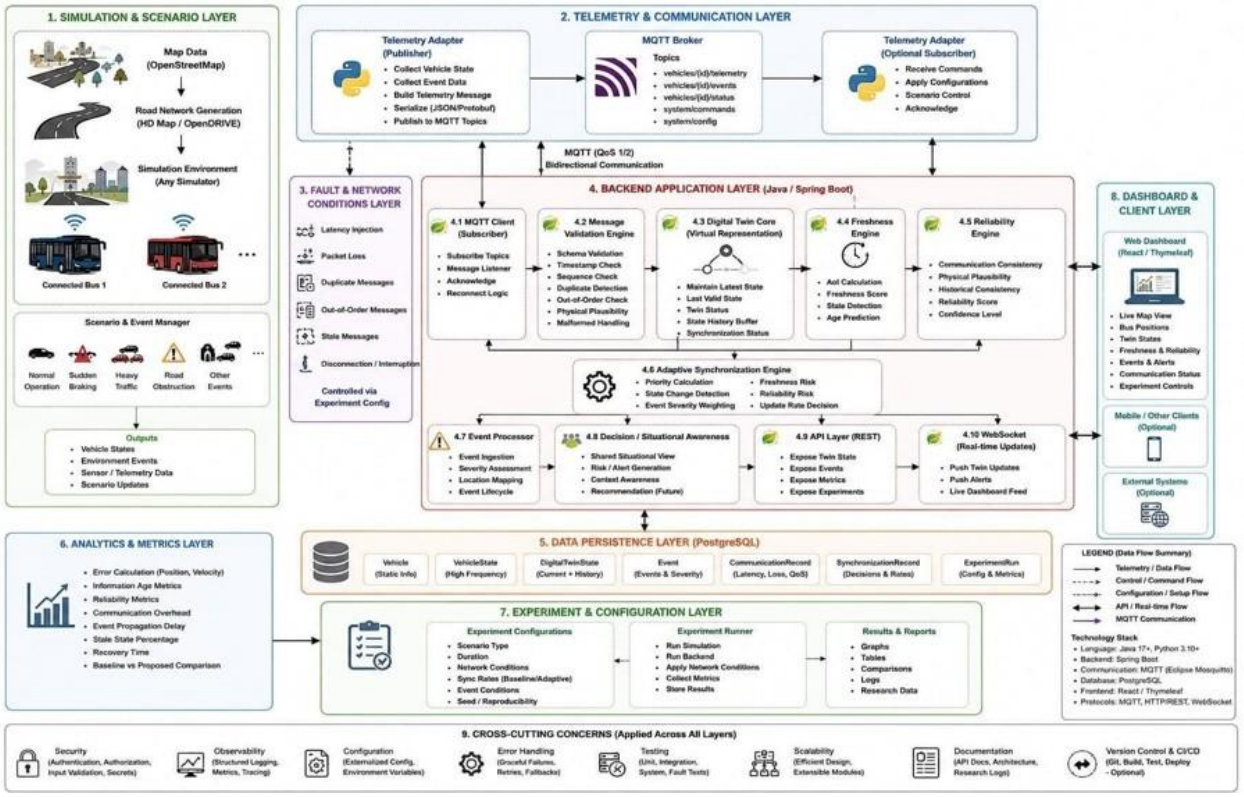


Figure 3.1 System Architecture

3.2 SYSTEM REQUIREMENTS

The software environment must support urban road simulation, vehicle-state generation, information exchange and Digital Twin representation. Python is used for implementation and integration of the project modules. OpenStreetMap is used as the source of urban road-network information and CARLA is used for vehicle simulation.

A computer with sufficient CPU, memory and graphics capability is required because CARLA is a real-time simulation platform. A dedicated GPU is recommended for smooth simulation and visualization. Internet connectivity may be required for obtaining map data or software packages when setting up the environment.

The requirements are intentionally limited to technologies supported by the current project scope. Advanced AI perception models, physical sensors and cloud services are not treated as implemented requirements unless separately integrated into the prototype.

- **Software:** Python, OpenStreetMap data, CARLA, operating system and development tools.
- **Hardware:** Computer or laptop, adequate RAM, suitable CPU, graphics capability and network connectivity.
- **Development:** Code editor, Python environment and version-control tools where applicablen

Software / Tool	Requirement	Purpose
Python	Python environment and required packages	Vehicle-state generation, integration and communication modules.
OpenStreetMap	Urban road-network data	Provides realistic road-network information for the simulated environment.
CARLA	CARLA simulator	Represents the urban environment and simulates the connected buses.
Operating System	Compatible desktop operating system	Hosts CARLA, Python tools and supporting development software.
Development Tools	Code editor and version-control tools where applicable	Supports implementation, testing and project maintenance.

Table 3.1 Software Requirements

Hardware	Recommended Requirement	Purpose
Computer / Laptop	System capable of running CARLA in real time	Runs simulation, development tools and monitoring components.
CPU	Adequate multi-core processor	Supports simulation, state generation and application processing.
RAM	Sufficient memory for CARLA and development tools	Maintains simulation and application processes without excessive memory pressure.
GPU	Dedicated or suitable graphics capability recommended	Supports smooth CARLA simulation and visualization.
Network Connectivity	Internet/network access when required	Supports map-data acquisition, package setup and future communication experiments.

Table 3.2 Hardware Requirements

Parameter	Description	Use in the Digital Twin
Vehicle Identifier	Unique identity of Bus 1 or Bus 2	Distinguishes the state of each connected bus.
Position	Current position in the simulated road network	Updates the virtual location of the bus.
Speed / Movement State	Current speed or movement condition	Represents the instantaneous movement state.
Route / Operational Status	Current route progress or operating status	Provides context for monitoring and coordination.
Timestamp / Update Sequence	Time or sequence associated with an update where available	Supports freshness, ordering and synchronization checks.

Table 3.3 Vehicle-State Parameters

3.3. DATA FLOW AND ACTIVITY DESIGN

The data flow begins with OpenStreetMap-based road information and the CARLA environment. The simulated buses generate vehicle-state information. The communication module exchanges relevant information between the buses, after which the shared information is provided to the Digital Twin. The Digital Twin maintains a virtual representation of the connected buses. Monitoring components consume the synchronized information to present the current transportation state and support situational awareness.

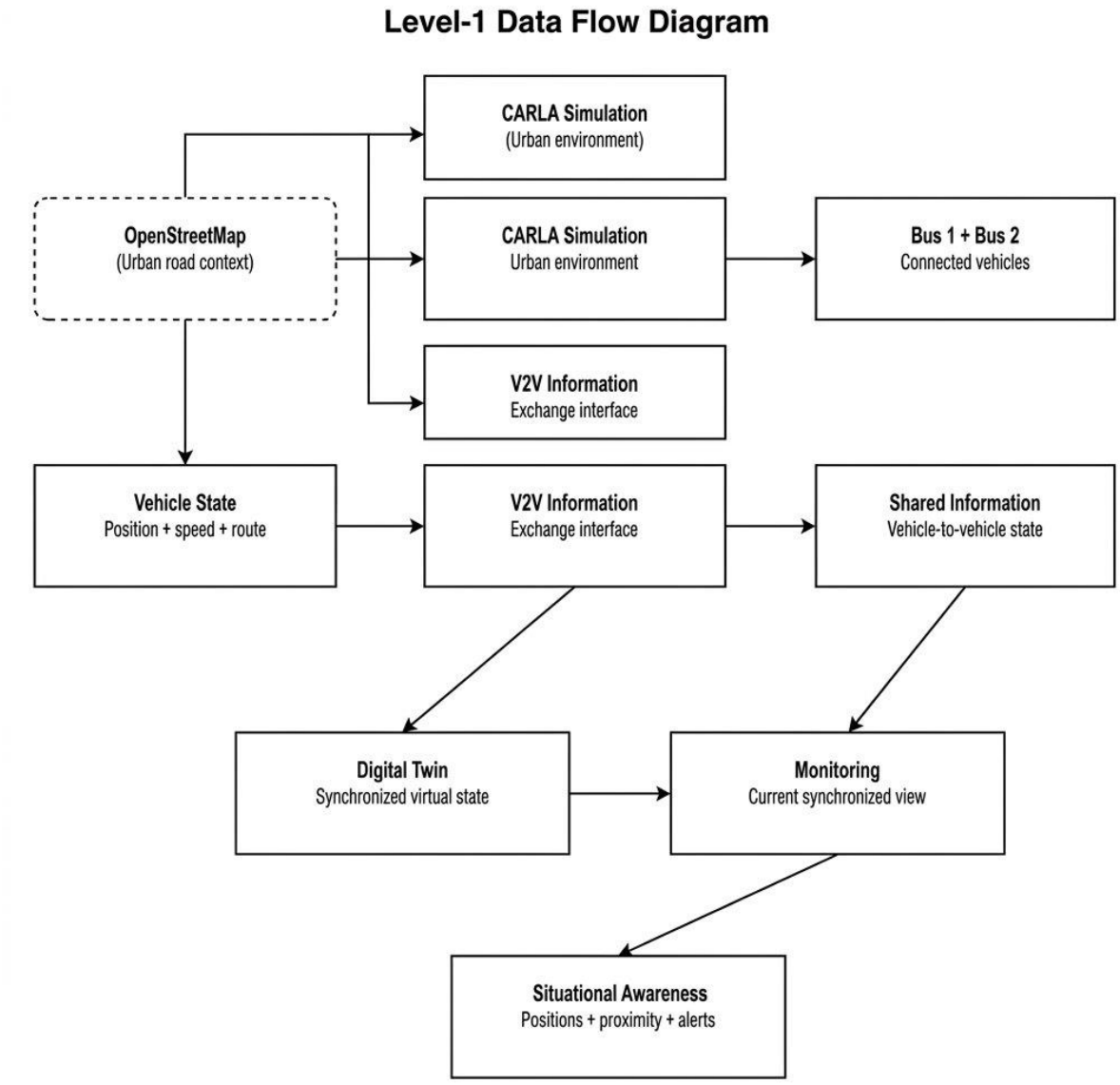


Figure 3.2 Level-1 Data Flow Diagram

The activity sequence starts by loading the urban environment and initializing CARLA. Two buses are then spawned. Their states are generated and updated as the simulation progresses.

The system exchanges vehicle information, updates the Digital Twin and displays the synchronized state. The monitoring stage allows the user to observe the current transportation environment. The loop continues as long as the simulation is active.

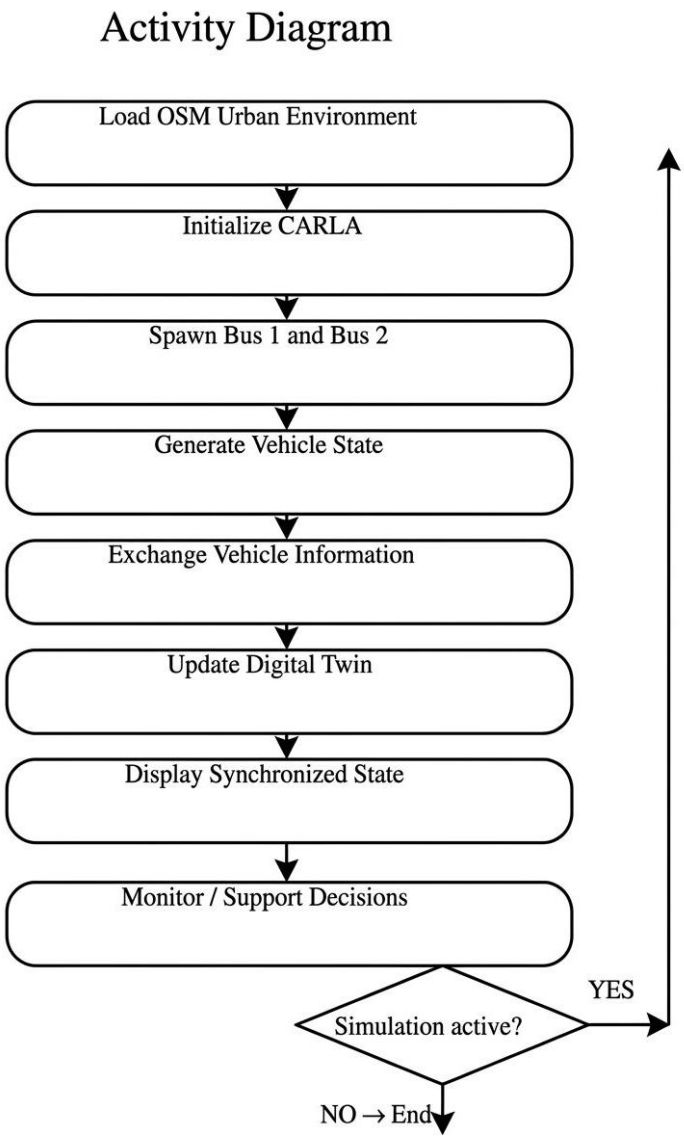


Figure 3.3 Activity Diagram

3.4 DEPLOYMENT AND SCALING

The current prototype is designed for local simulation. The deployment flow begins with the urban road environment and CARLA simulation, followed by Python-based vehicle-state and communication modules and the Digital Twin representation.

The architecture can be scaled by increasing the number of simulated buses, expanding the road network and introducing a dedicated communication service. A cloud-based Digital Twin can be considered when the system needs centralized monitoring across a larger fleet.

Future scaling should preserve low-latency state updates because transportation information is time-sensitive. Edge processing can be introduced for immediate vehicle-side operations, while cloud services can support long-term storage, analytics and fleet-level monitoring.

The present project does not claim large-scale fleet deployment, physical autonomous buses or full autonomous driving/control. These are identified as future extensions.

CHAPTER 4

PROJECT DESCRIPTION

The project is a simulation-oriented prototype that demonstrates how connected urban buses can share operational information and maintain a common virtual representation through a Digital Twin.

The methodology begins with realistic road-network modelling, continues with vehicle simulation and state generation, and then introduces cooperative information exchange and Digital Twin synchronization.

The project focuses on the system-level relationship among these components rather than implementing every advanced autonomous-driving capability. This makes it possible to validate the architecture in a controlled environment before moving toward more complex real-world deployments.

4.1 METHODOLOGIES

The following core modules define the simulation, communication, and synchronization pipeline for the connected transit system:

- Urban Environment Modelling: OpenStreetMap data provides realistic, real-world road-network layouts to build the simulation environment.
- Vehicle Simulation: CARLA generates the urban simulation landscape and controls the operations of two distinct buses within it.
- Vehicle State Generation: The system continuously tracks and outputs dynamic parameters for each bus, including position, speed, and route status.

- Vehicle-to-Vehicle (V2V) Information Exchange: A dedicated communication layer allows the simulated buses to mutually exchange operational data in real time.
- Digital Twin Synchronization: The Digital Twin platform ingests the streamed stream of data to keep the virtual representation of both buses fully updated.
- Situational Awareness: A live, synchronized view enables continuous monitoring and forms the foundational basis for transportation decision support.

4.2 MODULES

The prototype system relies on a modular architecture to handle environment loading, data generation, real-time communication, and synchronization:

- **Environment Module: Loads and manages** the OSM-based urban environment and handles its visual and structural representation inside **CARLA**.
- **Vehicle State Module:** Acts as a **live component** of the prototype that continuously generates and updates each bus's **position, speed, route status**, and active operational state.
- **Communication Module:** Provides the vital interface for **vehicle-to-vehicle (V2V) information exchange**, which is currently **mocked** to safely demonstrate the planned data flow.
- **Digital Twin Module:** Maintains the **virtual representations** of both Bus 1 and Bus 2, directly updating their properties using incoming stream data.
- **Monitoring and Decision Support Module:** Synthesizes and displays **synchronized information** to provide operator **situational awareness** and essential transit monitoring tools.

4.3 MODULE INTERACTION

The modules operate as a continuous pipeline. The environment module provides the simulated road context to CARLA. CARLA controls the simulated buses, while the vehicle-state module extracts or generates operational values.

The communication module makes the relevant information available to the other bus. The Digital Twin consumes the shared information and maintains the virtual state of the transportation environment.

Monitoring components then display this state.

The modular design is important because communication and perception capabilities can be enhanced independently in future versions. For example, the mocked communication interface can be replaced with a real V2V protocol without redesigning the Digital Twin module.

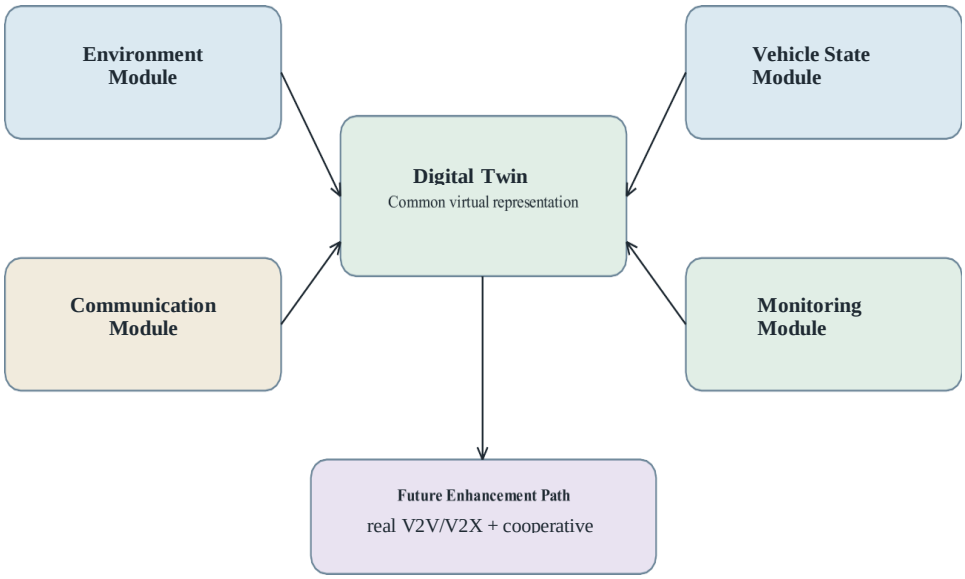


Figure 4.1 Module-level view of the integrated platform

CHAPTER 5

IMPLEMENTATION, RESULTS AND DISCUSSION

The implementation phase converts the proposed architecture into a working simulation prototype. The implementation is organized around the urban environment, vehicle simulation, vehicle-state generation, cooperative information exchange, Digital Twin representation and monitoring.

The prototype demonstrates the core connected-transportation workflow with two buses. Vehicle simulation, vehicle-state generation and Digital Twin representation are live components. The communication layer is currently represented through a mocked implementation so that the end-to-end information flow can be demonstrated.

The results are therefore evaluated primarily in terms of functional integration, state propagation, synchronization and monitoring rather than claiming full autonomous driving performance.

5.1 IMPLEMENTATIO RESULTS – URBAN ENVIRONMENT AND VEHICLE SIMULATION

The urban environment is prepared using OpenStreetMap road information and represented in CARLA. This provides a realistic road context for the two-bus scenario and allows the project to demonstrate vehicle movement within a structured transportation environment.

Two buses are spawned in the simulation and their movement is monitored. The simulation acts as the physical or operational layer of the Digital Twin, providing the state information that is later represented virtually.

The use of simulation makes it possible to repeat scenarios, observe state changes and validate the data flow without requiring physical autonomous buses.

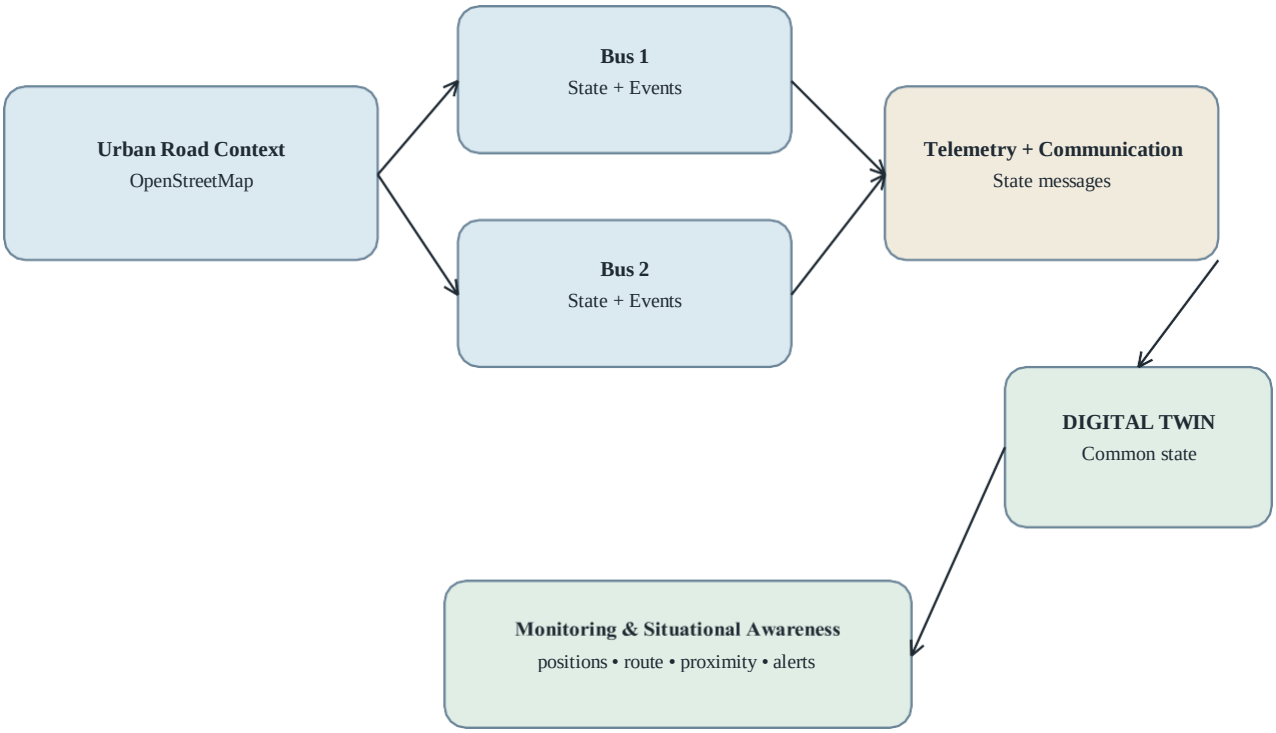


Figure 5.1 Urban environment, connected buses and Digital Twin pipeline

5.1.1 VEHICLE STATE GENERATION

The vehicle-state module is responsible for continuously generating the operational state of the two simulated buses. Position and speed provide the immediate movement state, while route status provides contextual information about the bus operation.

The state information forms the basic data unit exchanged between connected vehicles. Each update can be associated with the identity of the bus so that the Digital Twin can distinguish Bus 1 from Bus 2.

Continuous state generation is important because a Digital Twin becomes useful only when its virtual representation reflects changes in the underlying system. The live prototype demonstrates this continuous update concept.

- Vehicle identifier.
- Current position in the simulated road network.
- Current speed or movement state.
- Route or operational status.
- Timestamp or update sequence where available.

5.1.2 COOPERATIVE INFORMATION EXCHANGE

The cooperative communication module is designed to allow one bus to make relevant operational information available to another bus. The purpose is to demonstrate how connected vehicles can build awareness using information that is not generated locally.

In the current prototype, the communication mechanism is mocked. This means that the project demonstrates the planned message flow without claiming that a production-grade V2V radio or C-V2X stack has been implemented.

The prototype nevertheless establishes the interface required for future communication. A real implementation could use a V2V or V2X protocol and transmit structured messages containing vehicle identity, position, speed and route state.

5.1.3 DIGITAL TWIN IMPLEMENTATION

The Digital Twin module maintains a virtual representation of the two connected buses. The virtual model is updated using the vehicle-state information produced by the simulation and the information made available through the communication layer.

The Digital Twin provides a common representation instead of treating each bus as an isolated data source. This is useful for transportation monitoring because the state of multiple vehicles can be viewed together.

The implementation demonstrates the core concept of physical or simulated state being reflected in a virtual model. Future versions can enrich the Twin with road infrastructure, traffic events, passenger information and historical state.

5.1.4 MONITORING AND SITUATIONAL AWARENESS

The monitoring component presents the synchronized state of the connected buses. By observing the state of multiple vehicles together, the user can understand how the buses are positioned and operating within the simulated urban environment. The Digital Twin can therefore act as an intermediate layer between raw vehicle-state information and higher-level transportation decisions. For example, shared bus positions can support monitoring of vehicle proximity, route progress and possible coordination opportunities. The current project demonstrates basic situational awareness rather than automated control. The system does not claim to make safety-critical autonomous driving decisions.

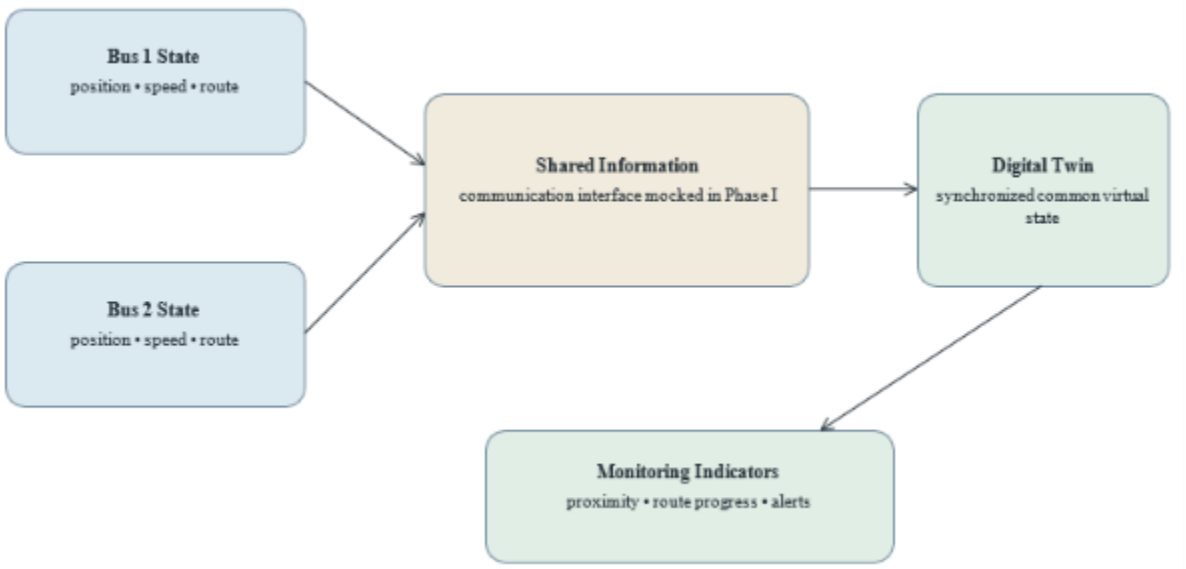


Figure 5.2 Information flow supporting monitoring and situational awareness

5.2 DISCUSSION

The prototype demonstrates the feasibility of integrating urban map data, vehicle simulation, vehicle-state generation, cooperative information exchange and Digital Twin representation in a single framework.

The most important outcome is the synchronized representation of two connected buses. The simulation generates live state information, the communication interface represents information sharing, and the Digital Twin provides a common virtual view.

The current implementation also clearly identifies the boundary between demonstrated functionality and future development. The mocked communication component should be replaced by an actual V2V or V2X implementation for a more complete cooperative perception experiment.

Performance evaluation in future work can include communication latency, state-update frequency, synchronization delay, scalability with increasing vehicle count and robustness under communication loss.

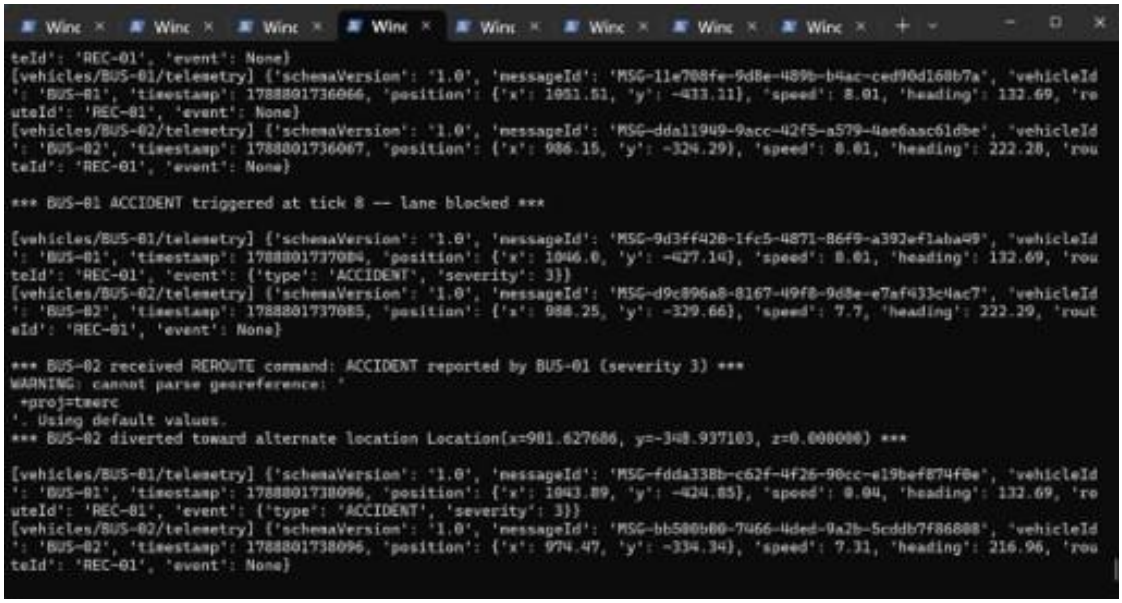


Figure 5.3 Vehicle telemetry and accident event log

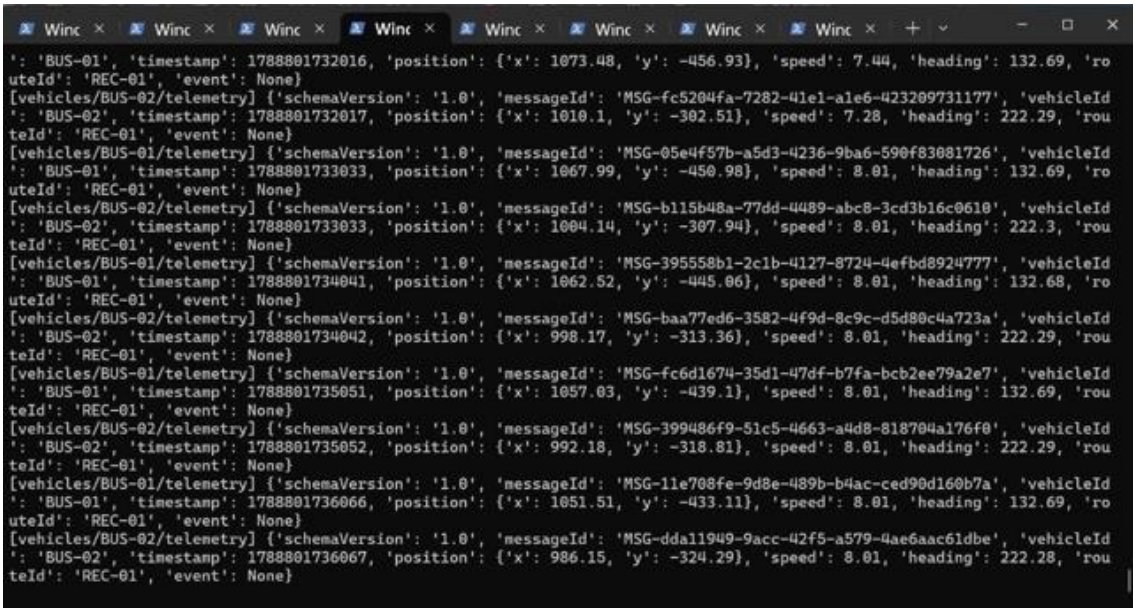


Figure 5.4 Vehicle telemetry stream for connected buses

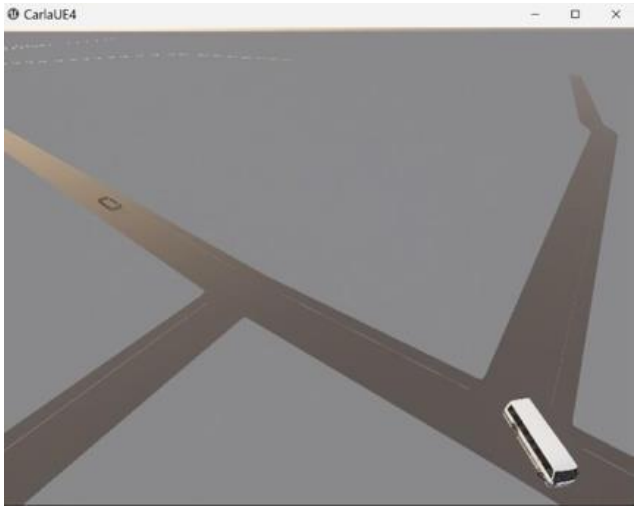


Figure 5.5 CARLA urban road environment – initial vehicle position

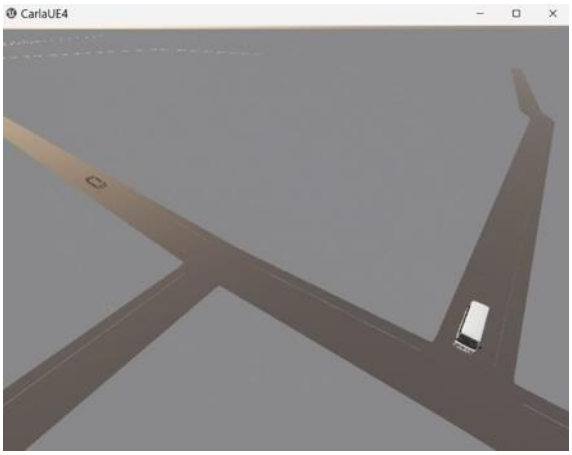


Figure 5.6 CARLA urban road environment – vehicle movement

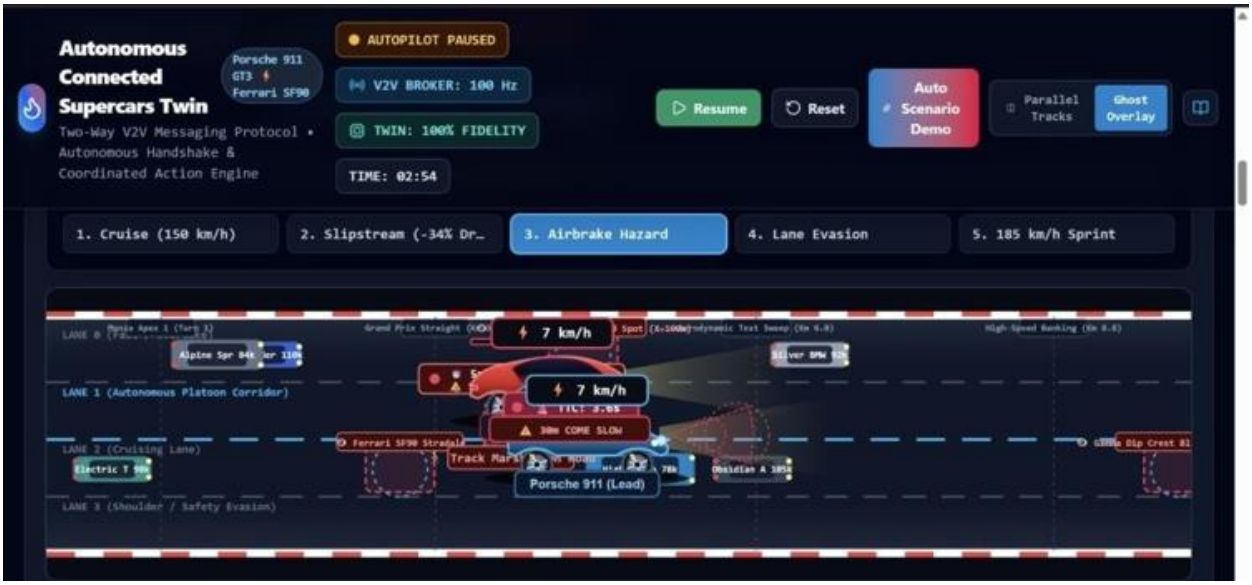


Figure 5.7 Connected Supercars Digital Twin – scenario monitoring dashboard



Figure 5.8 Digital Twin multi-angle perspective studio

5.3 TEST SCENARIOS AND PROTOTYPE STATUS

The prototype can be evaluated using repeatable scenarios involving two buses. A baseline scenario can verify that both buses are created and their states are updated. A communication scenario can verify that information from one bus becomes available to the other. A Digital Twin scenario can verify that the virtual state reflects the latest available information.

The following status represents the current Phase I prototype: vehicle simulation is live, vehicle-state generation is live, the communication module is mocked, the Digital Twin is live, and monitoring/situational awareness is demonstrated.

This status should be treated as the implementation baseline for the next project phase.

- **Urban environment:** Implemented.
- **CARLA simulation:** live.
- **Two buses:** live.
- **Vehicle-state generation:** live.
- **V2V information exchange:** mocked/prototype.

- **Digital Twin representation:** live.
- **Monitoring:** demonstrated.
- **Physical autonomous bus:** outside current scope.
- **Large-scale fleet:** outside current scope.
- **Full autonomous driving/control:** outside current scope.

Component	Current Status	Implementation / Scope
Urban environment	Implemented	OpenStreetMap-based road environment prepared for simulation.
CARLA simulation	Live	Two-bus urban simulation is operational.
Two buses	Live	Bus 1 and Bus 2 are represented in the simulation.
Vehicle-state generation	Live	Position, speed, route status and related state information are generated.
V2V information exchange	Mocked / Prototype	Planned information flow is demonstrated without claiming a production V2V/C-V2X stack.
Digital Twin representation	Live	Virtual state of the connected buses is maintained and updated.
Monitoring	Demonstrated	Synchronized state is presented for basic situational awareness.

Physical autonomous bus	Outside current scope	Reserved for future extension.
Large-scale fleet	Outside current scope	Architecture can be scaled in future phases.
Full autonomous driving/control	Outside current scope	Not claimed in the Phase I implementation.

Table 5.1 Prototype Implementation Status

Test Scenario	Condition / Input	Expected Behaviour	Current Status
Baseline simulation	Initialize CARLA with two buses	Both buses are created and their states are updated continuously.	Demonstrated
Vehicle-state update	Bus position/speed/route values change during simulation	Digital Twin reflects the latest available vehicle state.	Demonstrated
Information exchange	Operational information from one bus is made available to the other	Receiving side can access the shared vehicle information through the prototype interface.	Prototype / mocked
Digital Twin synchronization	New vehicle-state information reaches the Twin	Virtual representation updates to the latest available state.	Demonstrated

Monitoring	User observes synchronized bus states	Dashboard/monitoring view supports basic situational awareness.	Demonstrated
Communication limitation	Latency, packet loss or stale messages are introduced in an experiment configuration	System behaviour can be observed and future metrics such as delay and stale-state percentage can be evaluated.	Planned evaluation

Table 5.2 Test Scenarios and Expected Behaviour

The proposed AI-Driven Cooperative Perception and Digital Twin Platform presents a simulation-based framework for connected urban transportation. The system focuses on two buses operating within a realistic road environment created using OpenStreetMap and CARLA.

The framework generates vehicle information such as position, speed and route status and is designed to facilitate information exchange between connected buses. A Digital Twin maintains a synchronized virtual representation of the buses and their transportation environment.

The prototype successfully demonstrates vehicle simulation, vehicle-state generation and Digital Twin representation. The communication component is currently represented as a mocked prototype for demonstrating the planned vehicle-to-vehicle information exchange.

The resulting synchronized representation provides a foundation for monitoring, situational awareness and basic transportation decision support. The project therefore establishes a practical starting point for integrating cooperative transportation information and Digital Twin.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The project demonstrates that a connected transportation environment can be represented effectively through a combination of realistic map data, simulation, vehicle-state generation and Digital Twin technology.

The two-bus scenario provides a clear and manageable demonstration of cooperative information sharing. Instead of relying only on the local state of each vehicle, the framework creates a common view in which information from connected vehicles can be represented together.

CARLA provides the simulation environment, OpenStreetMap contributes realistic urban road information, the communication layer provides the planned mechanism for information exchange and the Digital Twin maintains the virtual representation.

Although the current prototype does not implement physical autonomous buses, large-scale fleet deployment or full autonomous driving and control, it establishes the architectural foundation required for these future extensions.

The project therefore achieves its Phase I objective of demonstrating a connected-vehicle and Digital Twin framework for an urban two-bus scenario.

6.2 FUTURE WORK

The next stage of development can extend the prototype in several directions. The most immediate enhancement is replacement of the mocked communication interface with an actual V2V or V2X communication protocol.

The system can then be extended from two buses to a larger fleet and tested under more complex urban traffic conditions. Additional road infrastructure and traffic entities can be represented in the Digital Twin to improve the completeness of the virtual environment.

Further work can introduce advanced cooperative perception using cameras, LiDAR, radar or other sensor modalities. AI-based object detection and sensor fusion can then be incorporated when the project scope requires perception-level information rather than only vehicle-state information.

A cloud-connected Digital Twin can support centralized fleet monitoring, while edge processing can be explored for low-latency vehicle-side operations. Future research can also investigate communication loss, synchronization delay, cybersecurity and privacy.

Physical vehicle integration and full autonomous driving/control remain longer-term extensions and are outside the current Phase I scope.

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APPENDICES

Appendix A — SOFTWARE AND HARDWARE REQUIREMENTS

Category	Item	Requirement / Specification	Purpose
Software	Python	Python environment and required packages	Vehicle-state generation, integration and communication modules
Software	OpenStreetMap	Urban road-network data	Provides realistic road-network information for the simulated environment
Software	CARLA	CARLA simulator	Represents the urban environment and simulates the connected buses
Software	Operating System	Compatible desktop operating system	Hosts CARLA, Python tools and supporting development software
Software	Development Tools	Code editor and version-control tools where applicable	Supports implementation, testing and project maintenance
Hardware	Computer / Laptop	System capable of running CARLA in real time	Runs simulation, development tools and monitoring components
Hardware	CPU	Adequate multi-core processor	Supports simulation, state generation and application processing
Hardware	RAM	Sufficient memory for CARLA and development tools	Maintains simulation and application processes
Hardware	GPU	Dedicated or suitable graphics capability recommended	Supports smooth CARLA simulation and visualization
Hardware	Network Connectivity	Internet/network access when required	Supports map-data acquisition, package setup and future communication experiments

Appendix B — CPU–GPU Benchmark Details

Parameter	Description	Use in Digital Twin
Vehicle Identifier	Unique identity of Bus 1 or Bus 2	Distinguishes the state of each connected bus

Position	Current position in the simulated road network	Updates the virtual location of the bus
Speed / Movement State	Current speed or movement condition	Represents the instantaneous movement state
Route / Operational Status	Current route progress or operating status	Provides context for monitoring and coordination
Timestamp / Update Sequence	Time or sequence associated with an update where available	Supports freshness, ordering and synchronization checks

APPENDIX C – SYSTEM MODULES AND FUNCTIONS

Module	Main Function	Current Phase I Status	Future Extension
Environment Module	Loads and manages the OSM-based urban environment and its representation in CARLA	Implemented	Expanded road infrastructure and traffic entities
Vehicle State Module	Generates and updates bus position, speed, route status and operational state	Live	More detailed vehicle and sensor states
Communication Module	Provides the interface for vehicle-to-vehicle information exchange	Mocked / Prototype	Actual V2V or V2X communication protocol
Digital Twin Module	Maintains virtual representations of Bus 1 and Bus 2 and updates their states	Live	Richer virtual environment and historical information
Monitoring and Decision Support Module	Displays synchronized information for situational awareness and transportation monitoring	Demonstrated	Advanced analytics and decision-support functions